Basic Algebra

Course Notes MA2205: Basic Algebra

Notes by Prof. Anirban Banerjee

Eklavya Raman

24MS024

March 11, 2026

Contents

Introduction

Around the year 1800, people already knew how to solve equations of the forms -

- Linear Equations: $ax + b = 0$
- Quadratic Equations: $ax^2 + bx + c = 0$
- Cubic Equations: $ax^3 + bx^2 + cx + d = 0$
- Quartic Equations: $ax^4 + bx^3 + cx^2 + dx + e = 0$

However, the general solution for equations of degree 5 or higher was not known. A young mathematician named Evariste Galois answered the question of the existence of a general solution for polynomial equations of degree 5 or higher, using a tool called groups.

Around the same time, Carl Friedrich Gauss introduced a technique called modular arithmetic, which laid the foundation for number theory. This technique shares a close relationship with groups.

The 1800s also saw a revolution in geometry. For over 2000 years, Euclidean Geometry was the only known geometry. However, mathematicians began to explore different geometries (specailly geometries on non-planar or curved surfaces). Groups became useful in studying these new geometries as well.

The formalization of groups by mathematicians supporting Galois and Gauss led to the birth of Group Theory and Abstract Algebra. Mathematicians realized that groups were very useful in studying various mathematical structures, and thus extended the idea of groups to other structures such as rings, fields, modules, vector spaces etc. It took years of effort by many mathematicians to formalize these structures and study their properties.

As time passed, the applications of these structures in various fields of mathematics and science became apparent. Today, it is used in computer science, cryptography, physics, chemistry, biology and many other fields.

This course aims to introduce the basic concepts of Group Theory, Ring Theory and some fundamental ideas in Abstract Algebra. We will explore the properties of these structures and their applications in various fields of mathematics and science.

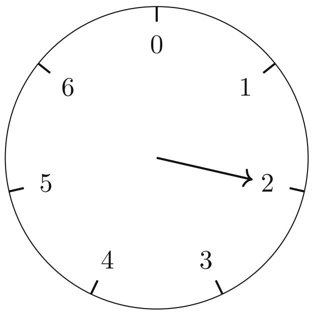
Chapter 1

Groups

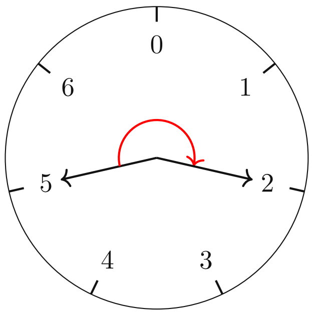
Before we formally define groups, we consider a few examples to get a basic idea of what groups are and how they arise in various mathematical contexts.

1.1 Examples of Groups

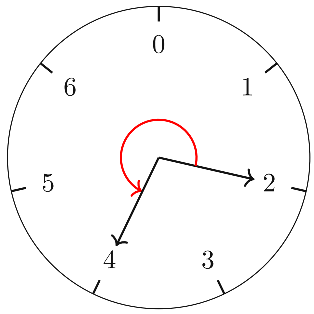
1. **Clock Arithmetic:** Consider a planet which spins faster than the Earth, such that a day on this planet is 7 hours long. The time on this planet can be represented using numbers from 0 to 6, where 0 represents midnight, 1 represents 1 AM, and so on. Let us visualise this using a clock with 7 hours.



   Let us define an operation of addition on this clock, where adding a number a to the current time b results in moving a hours forward on the clock. For example, if the current time is 5 and we add 4 hours, we move 4 hours forward on the clock, resulting in a time of 2 (since we wrap around after reaching 6). Let us visualise this operation on the clock.



   Here, we see that $5 + 4 = 2$. Similarly, we can define subtraction on this clock, where subtracting a number a from the current time b results in moving a hours backward on the clock. For example, if the current time is 2 and we subtract 5 hours, we move 5 hours backward on the clock, resulting in a time of 4 (since we wrap around after reaching 0). Let us visualise this operation on the clock.

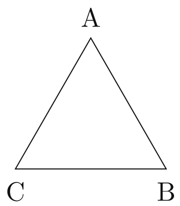


Notice that $2 - 5 = 4$, and also that subtraction can be thought of as adding the negative of a number, i.e., $2 - 5 = 2 + (-5) = 2 + 2 = 4$.

This type of arithmetic is called modular arithmetic, specifically arithmetic modulo 7 in this case. The set of numbers $\{0, 1, 2, 3, 4, 5, 6\}$ with the operations of addition and subtraction defined as above forms a mathematical structure known as a group. In this group, the operation of addition is closed, associative, has an identity element (0), and every element has an inverse (for example, the inverse of 5 is 2 since $5 + 2 \equiv 0 \mod 7$).

Modular arithmetic has applications in various fields such as cryptography, computer science, and number theory.

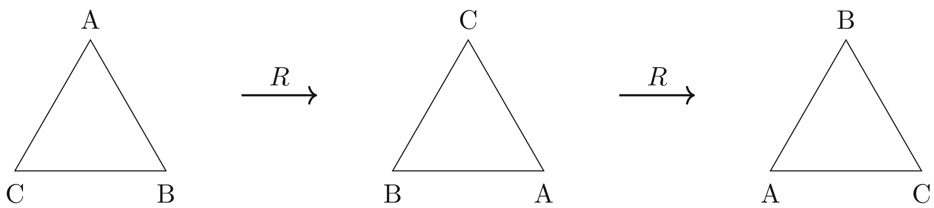
2. **Symmetries of an Equilateral Triangle:** Consider an equilateral triangle with vertices labeled A, B, C. The symmetries of this triangle include rotations and reflections that map the triangle onto itself. Let us visualize these symmetries.



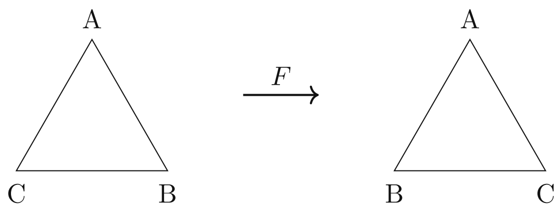
The operations that map the triangle onto itself are:

- **Rotations:** We can rotate the triangle about its center by 120° clockwise, to get the triangle mapped onto itself. [Denoted by R.]
- **Flipping:** Flipping the triangle about an axis passing through a vertex and the midpoint of the opposite side results in mapping the triangle onto itself. [Denoted by F.]
- **No Operation:** The triangle remains unchanged [Denoted by I].

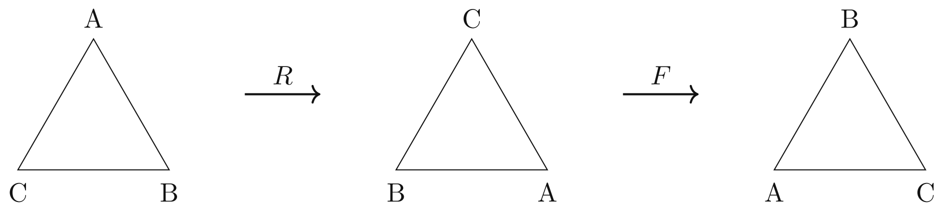
We can compose these fundamental operations to get all the symmetries of the triangle. Let us visualise these operations.



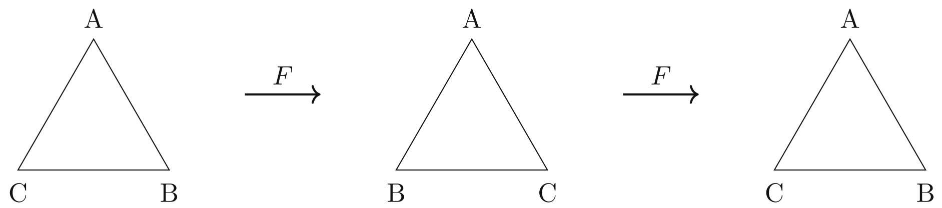
Similarly, we can visualize the flipping operation.

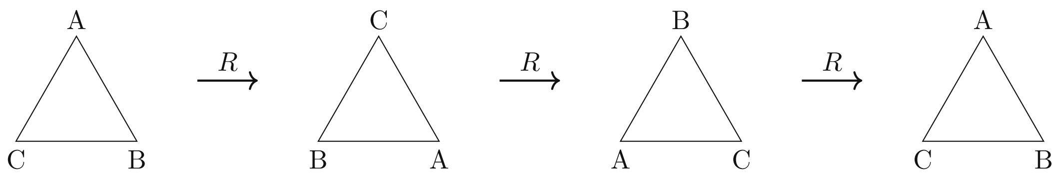


Let us now compose the flipping operation with rotations to get more symmetries.



Continuing this process, we find that there are a total of 6 symmetries of the equilateral triangle: I, R, R^2, F, FR, FR^2. Note that double flipping results in the identity operation, i.e., $F^2 = I$ and that $R^3 = I$. Let us visualise the above two operations to verify that they result in the identity operation.





3. **Integers under Addition:** Consider the set of integers $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ with the operation of addition.
   Note: We can define subtraction as adding the negative of a number, i.e., $x - y = x + (-y)$. We now have the following properties of this set equipped with the addition operation -
   - <u>Closure:</u> Let $x, y \in \mathbb{Z}$, then $x + y \in \mathbb{Z}$.
   - <u>Additive Identity:</u> There is a unique number $0 \in \mathbb{Z}$ such that for any $x \in \mathbb{Z}$, $x + 0 = x$. This number is called the identity element.

<u>Exercise 1.1:</u> Instead of an equilateral triangle, consider the follwing shapes and find all their symmetries -

- A square
- A rectangle (not a square)
- A cube
- A Tetrahedron

<u>Soln:</u> Let us consider just the case of a square here.
Square: Consider a square with vertices labeled A, B, C, D. We can do the operations of rotation by $90°$ clockwise (denoted by R), Flipping about the vertical axis (F_v), flipping about diagonal axis (F_d) and no operation (I). We can compose these operations to get all the symmetries of the square. There are a total of 8 symmetries of the square.
Similarly, we can find the symmetries of the other shapes.

Let us now analyze the similarities between these examples.

1. Each of these examples has a set of objects that are considered (numbers of the clock, symmetries of the triangle, integers). We call these objects elements of a set.

2. We have a way of combining two elements of the set to get another element of the set (addition on the clock, composition of symmetries, addition of integers). We call this way of combining elements an operation.

3. All sets are closed under the operation defined on them (adding two numbers on the clock gives another number on the clock, composing two symmetries gives another symmetry, adding two integers gives another integer).

4. The operation is associative (for any three numbers on the clock, adding them in any order gives the same result, composing three symmetries in any order gives the same result, adding three integers in any order gives the same result).

5. There is an identity element in each set with respect to the operation (0 on the clock, identity symmetry of the triangle, 0 in integers).

6. Every element has an inverse with respect to the operation (for any number on the clock, there is another number that when added gives 0, for any symmetry of the triangle, there is another symmetry that when composed gives the identity symmetry, for any integer, there is another integer that when added gives 0).

1.2 Groups and Operations

1.2.1 Definitions

Based on the observations from the previous section, we can now formally define groups and operations.

Definition 1.2.1.1

Groups
A group $(G, *)$ is a set G equipped with a ***binary operation*** $* : G \times G \to G$ that satisfies the following properties:

- **Closure:** For all $a, b \in G$, the result of the operation $a * b$ is also in G.
- **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
- **Identity Element:** There exists an element $e \in G$ such that for every element $a \in G$, $e * a = a * e = a$.
- **Inverse Element:** For each $a \in G$, there exists an element $b \in G$ such that $a*b = b*a = e$, where e is the identity element.

It took years of effort to develop the formal definition of groups based on the most fundamental properties observed in various mathematical contexts. For example, the property of commutativity (i.e., $a * b = b * a$ for all $a, b \in G$) is not included in the definition of groups, as there exist many important groups where this property does not hold. Groups where commutativity holds are called abelian groups, named after the mathematician Niels Henrik Abel. Similarly, a non-commutative group is called non-abelian.
Let us now formally define a binary operation.

Definition 1.2.1.2

Binary Operation
A binary operation on a set S is a function $* : S \times S \to S$ that combines any two elements $a, b \in S$ to produce another element $c \in S$, denoted as $a * b = c$.

Let us consider some examples of binary operations.

1. **Addition on Integers:** The operation of addition $+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ defined by $(a, b) \mapsto a + b$ is a binary operation on the set of integers $\mathbb{Z}$.

2. **Composition of Symmetries:** Consider a set X equipped with some structure. Now, a symmetry of X is a bijective function from X to itself that preserves the structure of X. The operation of composition of symmetries $\circ : \mathrm{Sym}(X) \times \mathrm{Sym}(X) \to \mathrm{Sym}(X)$ defined by $(f, g) \mapsto f \circ g$ is a binary operation on the set of symmetries $\mathrm{Sym}(X)$.

3. **Union and Intersection:** Let S be a non-empty set. Then, $\cup$ and $\cap$ are binary operations on the power set of S, denoted by $\mathcal{P}(S)$. We define the operations as follows:

$$\cup : \mathcal{P}(S) \times \mathcal{P}(S) \to \mathcal{P}(S)$$

defined by $(A, B) \mapsto A \cup B$

$$\cap : \mathcal{P}(S) \times \mathcal{P}(S) \to \mathcal{P}(S)$$

defined by $(A, B) \mapsto A \cap B$

4. **Matrix Multiplication:**For $n \in \mathbb{N}, n \geq 1$, let $GL_n(\mathbb{R})$ be the set of all invertible $n \times n$ matrices with real entries. Then, the operation of matrix multiplication $\cdot : GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \to GL_n(\mathbb{R})$ defined by $(A, B) \mapsto A \cdot B$ is a binary operation on the set of invertible matrices $GL_n(\mathbb{R})$.

Exercise 1.2: Is the matrix multiplication operation associative and commutative on the set $GL_n(\mathbb{R})$?
Soln: The operation of matrix multiplication is associative but not commutative. That is, for any three matrices $A, B, C \in GL_n(\mathbb{R})$, we have $(A \cdot B) \cdot C = A \cdot (B \cdot C)$, but in general, $A \cdot B \neq B \cdot A$.
Exercise 1.3: which of the following are binary operations on $\mathbb{N}$?

1. $(a, b) \mapsto a - b$

2. $(a, b) \mapsto \pm\sqrt{ab}$

3. $(a, b) \mapsto 5$

Soln: Let us analyze each operation:

1. The operation $(a, b) \mapsto a - b$ is not a binary operation on $\mathbb{N}$ because the result of the operation may not be a natural number. For example, if $a = 3$ and $b = 5$, then $a - b = -2$, which is not in $\mathbb{N}$.

2. The operation $(a, b) \mapsto \pm\sqrt{ab}$ is not a binary operation on $\mathbb{N}$ because the result of the operation may not be a natural number. For example, if $a = 2$ and $b = 8$, then $\sqrt{ab} = \sqrt{16} = 4$, which is in $\mathbb{N}$, but the negative root -4 is not in $\mathbb{N}$.

3. The operation $(a, b) \mapsto 5$ is a binary operation on $\mathbb{N}$ because for any natural numbers a, b, the result of the operation is always 5, which is in $\mathbb{N}$.

Exercise 1.4: Is the operation on $\mathbb{R}$ defined by $(a, b) \mapsto \frac{a-b}{2}$ commutative and associative?
Soln: Let us analyze the operation $(a, b) \mapsto \frac{a-b}{2}$:

- **Commutativity:** The operation is not commutative because in general, $\frac{a-b}{2} \neq \frac{b-a}{2}$. For example, if $a = 3$ and $b = 5$, then $\frac{3-5}{2} = -1$ and $\frac{5-3}{2} = 1$, which are not equal.

- **Associativity:** The operation is not associative because in general, $\frac{\left(\frac{a-b}{2}\right)-c}{2} \neq \frac{a-\left(\frac{b-c}{2}\right)}{2}$. For example, if $a = 4, b = 2, c = 0$, then:

$$\frac{\left(\frac{4-2}{2}\right) - 0}{2} = \frac{1 - 0}{2} = 0.5$$

and

$$\frac{4 - \left(\frac{2-0}{2}\right)}{2} = \frac{4 - 1}{2} = 1.5$$

which are not equal.

1.2.2 Cayley Tables

Let S be a non-empty set and $*$ be a binary operation on S. We can represent the operation $*$ using a table called the Cayley table. The rows and columns of the table are labeled with the elements of S, and the entry in the row corresponding to element a and the column corresponding to element b is the result of the operation $a * b$.
Let us consider the following example -
Let S $= \{\pm 1, \pm i\}$, where i is the imaginary unit. We define a binary operation $*$ on S as follows -

$*$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

Note 1.2.2.1

We note the following properties of the above Cayley table -

1. We have 1 as the identity element, since $1 * a = a * 1 = a$ for all $a \in S$.
2. Every row and every colum contain the identity element, $1 \cdot 1 = 1, -1 \cdot -1 = 1, i \cdot -i = 1, -i \cdot i = 1$
3. Asssociativity holds, i.e., $(a * b) * c = a * (b * c)$ for all $a, b, c \in S$.
4. The table is symmetric about the main diagonal, i.e., $a * b = b * a$ for all $a, b \in S$. Thus, the operation is commutative.
5. No duplicate elements appear in any row or column. If there existed duplicates, then there would be elements without inverses.

Based on the above properties, we can conclude that the set S with the operation $*$ forms an abelian group.

We now consider a few exercises based on Cayley tables.
Exercise 1.5: Consider the group $(\{e, a, b\}, *)$. Find the cayley table for this group, given that e is the identity element, and that $a * a = b, a * b = e, b * a = e, b * b = a$. Show that the group is abelian.
Soln: We can construct the Cayley table as follows -

$*$	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

To show that the group is abelian, we need to verify that the operation is commutative, i.e., $a * b = b * a$ for all $a, b \in \{e, a, b\}$. We can see that the cayley table is symmetric about the main diagonal, which implies that the operation is commutative. Therefore, the group is abelian.

Exercise 1.6: Using Cayley tables, find all the groups of order 4 (i.e., groups with 4 elements). Hint: There are 4 possible tables, 3 are identical to each other, and 2 distinct groups exist. Soln: Let us denote the elements of the group as $\{e, a, b, c\}$, where e is the identity element. We can construct the Cayley tables for the possible groups of order 4.
Group 1:

$*$	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

Group 2:

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

These are the two distinct groups of order 4. The other two possible tables are identical to one of these two groups. Thus, we have found all the groups of order 4 using Cayley tables.

Exercise 1.7: Let $(G, *)$ be a group. Prove the following -

1. The identity element is unique.
2. For each element $a \in G, \exists$ unique $b \in G$ such that $a * b = b * a = e$, where e is the identity element.
3. For all $a, b, c \in G$, if $a * b = a * c$, then $b = c$ (Left Cancellation Law). Similarly, if $b * a = c * a$, then $b = c$ (Right Cancellation Law).
4. For all $a, b \in G$, the equation $a * x = b$ has a unique solution for $x \in G$. Similarly, the equation $y * a = b$ has a unique solution for $y \in G$.
5. $(a^{-1})^{-1} = a$ for all $a \in G$ and $(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$.
6. $\forall a \in G, m, n \in \mathbb{Z}, a^m * a^n = a^{m+n}$, where a^m denotes the m-th power of a with respect to the operation $*$. And, $(a^m)^n = a^{mn}$.

Soln: Let us prove each of the statements one by one -

1. Suppose there are more than 1 distinct identity elements in the group, say e_1 and e_2. By the definition of identity element, we have $e_1 * a = a$ and $e_2 * a = a$ for all $a \in G$. In particular, for $a = e_2$, we have $e_1 * e_2 = e_2$. Similarly, for $a = e_1$, we have $e_2 * e_1 = e_1$. Thus, we have $e_1 = e_2$, which contradicts our assumption that they are distinct. Therefore, the identity element is unique.
2. Let $a \in G$ be an arbitrary element. By the definition of inverse element, there exists an element $b \in G$ such that $a * b = e$ and $b * a = e$, where e is the identity element. Suppose there exists another element $c \in G$ such that $a * c = e$ and $c * a = e$. Then, we have:
$$b = b * e = b * (a * c) = (b * a) * c = e * c = c$$
Thus, $b = c$, which shows that the inverse element is unique.
3. Let $a, b, c \in G$ such that $a * b = a * c$. We want to show that $b = c$. By the definition of inverse element, there exists an element $a^{-1} \in G$ such that $a^{-1} * a = e$, where e is the identity element. Now, we have:
$$a^{-1} * (a * b) = a^{-1} * (a * c) \implies (a^{-1} * a) * b = (a^{-1} * a) * c \implies e * b = e * c \implies b = c$$
The same logic can be applied to prove the right cancellation law.
4. We have $a, b \in G$. We want to show that the equation $a * x = b$ has a unique solution for $x \in G$. By the definition of inverse element, there exists an element $a^{-1} \in G$ such that $a^{-1} * a = e$, where e is the identity element. Now, we can multiply both sides of the equation $a * x = b$ by a^{-1} from the left:
$$a^{-1} * (a * x) = a^{-1} * b \implies (a^{-1} * a) * x = a^{-1} * b \implies e * x = a^{-1} * b \implies x = a^{-1} * b$$
Thus, the equation $a * x = b$ has a unique solution for $x \in G$. Similarly, we can show that the equation $y * a = b$ has a unique solution for $y \in G$.
5. We know that every $a \in G$ has a unique inverse element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$, where e is the identity element. From this, we can deduce that $(a^{-1})^{-1} = a$ for all $a \in G$.
Now, let $a, b \in G$. We want to show that $(a*b)^{-1} = b^{-1}*a^{-1}$. By the definition of inverse element, we have:
$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * e * a^{-1} = a * a^{-1} = e$$
Similarly, we can show that $(b^{-1}*a^{-1})*(a*b) = e$. Thus, we have shown that $(a*b)^{-1} = b^{-1}*a^{-1}$ for all $a, b \in G$.

6. We prove the properties using mathematical induction.
   Base Case: For $n = 1$, we have $a^m * a^1 = a^{m+1}$ and $(a^m)^1 = a^{m \cdot 1} = a^m$, which holds true.
   Inductive Step: Assume that the properties hold for some $n = k$, i.e., $a^m * a^k = a^{m+k}$ and $(a^m)^k = a^{m \cdot k}$. We need to show that the properties hold for $n = k + 1$.
   We have:
   $$a^m * a^{k+1} = a^m * (a^k * a) = (a^m * a^k) * a = a^{m+k} * a = a^{m+k+1}$$
   and
   $$(a^m)^{k+1} = (a^m)^k * a^m = a^{m \cdot k} * a^m = a^{m \cdot k+m} = a^{m \cdot (k+1)}$$
   Thus, by the principle of mathematical induction, the properties hold for all $n \in \mathbb{Z}$.

We now give some examples of infinite non-abelian groups.
Example 1.2.2.1 -
$GL_n(\mathbb{R})$, the group of all invertible $n \times n$ matrices with real entries under the operation of matrix multiplication, is an infinite non-abelian group for $n \geq 2$. The identity element in this group is the identity matrix, and the inverse of a matrix is its inverse matrix. The operation of matrix multiplication is associative but not commutative, which makes $GL_n(\mathbb{R})$ a non-abelian group.
Similarly $SL_n(\mathbb{R})$, the group of all $n \times n$ matrices with real entries and determinant equal to 1 under the operation of matrix multiplication, is also an infinite non-abelian group for $n \geq 2$. The identity element in this group is also the identity matrix, and the inverse of a matrix is its inverse matrix. The operation of matrix multiplication is associative but not commutative, which makes $SL_n(\mathbb{R})$ a non-abelian group.
In general, for any infinite field F and any $n \geq 2$, the groups $GL_n(F)$ and $SL_n(F)$ are infinite non-abelian groups under the operation of matrix multiplication.
Example 1.2.2.2 -
$(\mathbb{Z}/n\mathbb{Z}, +)$, the group of integers modulo n under the operation of addition, is a finite abelian group. The elements of this group are the equivalence classes of integers modulo n, denoted by $[0], [1], [2], \ldots, [n-1]$. The identity element in this group is $[0]$, and the inverse of an element $[a]$ is $[-a]$. The operation of addition is associative and commutative, which makes $\mathbb{Z}/n\mathbb{Z}$ an abelian group.

Note 1.2.2.2

Each element of $\mathbb{Z}/n\mathbb{Z}$ is an equivalence class of integers. For example, in $\mathbb{Z}/5\mathbb{Z}$, we have the following equivalence classes -

$$[0] = \{\ldots - 10, -5, 0, 5, 10, \ldots\} \tag{1.2.2.1}$$
$$[1] = \{\ldots - 9, -4, 1, 6, 11, \ldots\} \tag{1.2.2.2}$$
$$[2] = \{\ldots - 8, -3, 2, 7, 12, \ldots\} \tag{1.2.2.3}$$
$$[3] = \{\ldots - 7, -2, 3, 8, 13, \ldots\} \tag{1.2.2.4}$$
$$[4] = \{\ldots - 6, -1, 4, 9, 14, \ldots\} \tag{1.2.2.5}$$

Therefore, each element of $\mathbb{Z}/n\mathbb{Z}$ is itself a countably infinite subset of integers. To make sense of the operation of addition on $\mathbb{Z}/n\mathbb{Z}$, we define the addition of two equivalence classes as follows -

$$[a] + [b] = [a + b]$$

Now, another interesting fact is that $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0] = \{[1], [2], \ldots, [n-1]\}$ forms a group under the operation of multiplication for the case of $n = 5$. The identity element in this group is $[1]$, and the inverse of an element $[a]$ is $[a^{-1}]$, where a^{-1} is the multiplicative inverse of a modulo n. The operation of multiplication is associative and commutative, which makes $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ an abelian group.

Note 1.2.2.3

Here, we define multiplication as follows -

$$[a] \cdot [b] = [a \cdot b]$$

where $a \cdot b$ is the product of a and b modulo n. Note that the operation of multiplication is only defined for the non-zero elements of $\mathbb{Z}/n\mathbb{Z}$, since $[0]$ does not have a multiplicative inverse. Therefore, we consider the set $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ when defining the group under multiplication.

We also note, that the same is not a group if n is not prime. We state the following result and prove it -

Theorem 1.1. *The group $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ under multiplication is a group if and only if n is prime.*

Proof. $(\rightarrow)$Suppose n is not prime and that $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ under multiplication is a group. Since n is not prime, there exist integers a and b such that $1 < a < n$, $1 < b < n$ (as every non prime posseses a unique prime factorization.), and $n = a \cdot b$. Consider the elements $[a]$ and $[b]$ in $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$. We have:

$$[a] \cdot [b] = [a \cdot b] = [n] = [0]$$

This contradicts the fact that $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ is a group under multiplication, since the product of two non-zero elements cannot be zero in a group. Therefore, if $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ is a group under multiplication, then n must be prime.
$(\leftarrow)$ Suppose n is prime. We need to show that $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ under multiplication is a group. The identity element in this group is $[1]$, since $[1]\cdot[a] = [a]\cdot[1] = [a]$ for all $[a] \in \{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$. Now, let $[a] \in \{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$. Since n is prime, there exists an integer a^{-1} such that $a \cdot a^{-1} \equiv 1 \mod n$ (we prove this seperately). Therefore, we have:

$$[a] \cdot [a^{-1}] = [a \cdot a^{-1}] = [1]$$

Thus, every element in $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ has a multiplicative inverse. The operation of multiplication is associative and commutative, which makes $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ an abelian group under multiplication. Therefore, if n is prime, then $\{\mathbb{Z}/n\mathbb{Z}\}\backslash[0]$ under multiplication is a group. □

Note 1.2.2.4

To show that if n is prime, there exists an integer a^{-1} such that $a \cdot a^{-1} \equiv 1 \mod n$, consider $a \in \mathbb{Z}, 1 < a < n$. Since n is prime, the greatest common divisor of a and n is 1. We claim there exists integers x and y such that $ax + ny = 1$. This can be shown using the Euclidean algorithm as follows -

$$\begin{aligned}
n &= q_1 a + r_1 \\
a &= q_2 r_1 + r_2 \\
r_1 &= q_3 r_2 + r_3 \\
&\vdots \\
r_{k-2} &= q_k r_{k-1} + r_k \\
r_{k-1} &= q_{k+1} r_k + 0
\end{aligned}$$

where q_i are the quotients and r_i are the remainders. Since the greatest common divisor of a and n is 1, we have $r_k = 1$. Now, we can express 1 as a linear combination of a and n by back-substituting the equations above:

$$\begin{aligned}
1 &= r_k \\
&= r_{k-2} - q_k r_{k-1} \\
&= r_{k-2} - q_k(r_{k-3} - q_{k-1} r_{k-2}) \\
&= (1 + q_k q_{k-1}) r_{k-2} - q_k r_{k-3} \\
&\vdots \\
&= xa + yn
\end{aligned}$$

Thus, we have shown that there exist integers x and y such that $ax+ny=1$. Taking this equation modulo n, we get $ax \equiv 1 \mod n$. Therefore, we can take $a^{-1}=x$, which shows that there exists an integer a^{-1} such that $a \cdot a^{-1} \equiv 1 \mod n$.

Example 1.2.2.3 -
Let $A \subseteq \mathbb{R}$ be a non-empty set. The set $C(A)$ of all continous functions from A to $\mathbb{R}$ forms a group under the operation of function addition, defined by $(f+g)(x) = f(x)+g(x)$ for all $f, g \in C(A)$ and $x \in A$. The identity element in this group is the zero function, defined by $0(x) = 0$ for all $x \in A$. The inverse of a function $f \in C(A)$ is the function $-f$, defined by $(-f)(x) = -f(x)$ for all $x \in A$. The operation of function addition is associative and commutative, which makes $C(A)$ an abelian group under function addition.

1.2.3 Subgroups

Definition 1.2.3.1

Subgroup
Let $(G, *)$ be a group. A subset H of G is called a subgroup of G if H itself forms a group under the operation $*$ inherited from G. In other words, $(H, *)$ is a subgroup of $(G, *)$ if the following conditions hold:

- **Closure:** For all $a, b \in H$, the result of the operation $a * b$ is also in H.
- **Identity Element:** The identity element of G is also in H.
- **Inverse Element:** For each element $a \in H$, its inverse with respect to the operation $*$ in G is also in H.

We now discuss finite subgroups of $(C^x, \cdot)$, the group of non-zero complex numbers under multiplication. Let n be a positive integer. Consider the set $H_n = \{e^{2\pi ik/n} : k = 0, 1, 2, \ldots, n-1\}$. We can verify that H_n is a subgroup of $(C^x, \cdot)$ as follows:

- **Closure:** For any two elements $e^{2\pi ik/n}$ and $e^{2\pi im/n}$ in H_n, their product is given by:

  $$e^{2\pi ik/n} \cdot e^{2\pi im/n} = e^{2\pi i(k+m)/n}$$

  Since $k+m$ is an integer, we can write it as $k+m = qn+r$ for some integer q and $0 \leq r < n$. Thus, we have:

  $$e^{2\pi i(k+m)/n} = e^{2\pi i(qn+r)/n} = e^{2\pi iq} \cdot e^{2\pi ir/n} = e^{2\pi ir/n}$$

  Since $0 \leq r < n$, we have $e^{2\pi ir/n} \in H_n$. Therefore, the product of any two elements in H_n is also in H_n, which shows that H_n is closed under multiplication.

- **Identity Element:** The identity element in $(C^x, \cdot)$ is the complex number 1, which can be expressed as $e^{2\pi i \cdot 0/n}$. Since 0 is an integer, we have $e^{2\pi i \cdot 0/n} = 1 \in H_n$. Thus, the identity element of $(C^x, \cdot)$ is also in H_n.

- **Inverse Element:** For each element $e^{2\pi ik/n}$ in H_n, its inverse with respect to multiplication is given by:

  $$(e^{2\pi ik/n})^{-1} = e^{-2\pi ik/n}$$

  We can express this as:

  $$e^{-2\pi ik/n} = e^{2\pi i(n-k)/n}$$

  Since $n-k$ is an integer and $0 < n-k < n$, we have $e^{2\pi i(n-k)/n} \in H_n$. Therefore, the inverse of each element in H_n is also in H_n. Thus, H_n satisfies the conditions for being a subgroup of $(C^x, \cdot)$.

1.3 Cyclic Groups

Definition 1.3.0.1

Cyclic Group
A group $(G, *)$ is called a cyclic group if there exists an element $g \in G$ such that every element of G can be expressed as a power of g (with respect to the operation $*$). In other words, $G = \{g^n : n \in \mathbb{Z}\}$, where g^n denotes the n-th power of g with respect to the operation $*$. The element g is called a generator of the cyclic group G.

We can have both finite and infinite cyclic groups under certain conditions. Consider the following cases -

1. ***Powers of g are distinct:*** If the powers of g are distinct, i.e., $g^m \neq g^n$ for all $m, n \in \mathbb{Z}$ with $m \neq n$, then the cyclic group $G = \{g^n : n \in \mathbb{Z}\}$ is infinite. In this case, the elements of G are all distinct and there are countably infinite elements in G (since there is a bijection between G and $\mathbb{Z}$ given by $g^n \leftrightarrow n$).

2. ***Powers of g are not distinct:*** If there exist integers $m, n \in \mathbb{Z}$ with $m \neq n$ such that $g^m = g^n$, then the cyclic group $G = \{g^n : n \in \mathbb{Z}\}$ is finite. In this case, there exists a positive integer k such that $g^k = e$, where e is the identity element of the group. The order of the cyclic group G is equal to the smallest positive integer k such that $g^k = e$. Note that this set has a bijection with the set of integers modulo k, denoted by $\mathbb{Z}/k\mathbb{Z}$, which is a finite cyclic group of order k.

3. ***Trivial Case:*** If $g = e$, where e is the identity element of the group, then the cyclic group $G = \{g^n : n \in \mathbb{Z}\}$ is trivial and contains only the identity element. In this case, $G = \{e\}$ and it is a finite cyclic group of order 1.

Note 1.3.0.1

We denote a cyclic group of order n by C_n and a cyclic group of infinite order by C_∞. Thus, we have the following classifications of cyclic groups:

- If G is a cyclic group of order n, then $G \cong C_n \cong \mathbb{Z}/n\mathbb{Z}$.
- If G is a cyclic group of infinite order, then $G \cong C_\infty \cong \mathbb{Z}$.
- If G is a trivial cyclic group, then $G \cong C_1 \cong \{e\}$.

We now define the order of an element in a group.

Definition 1.3.0.2

Order of an Element
Let $(G, *)$ be a group and let $g \in G$ be an element of the group. The order of the element g, denoted by $\operatorname{ord}(g)$, is defined as the smallest positive integer k such that $g^k = e$, where e is the identity element of the group. If no such positive integer exists, we say that g has infinite order.

Example 1.3.0.1 **-**
Consider the group $(\mathbb{Z}/6\mathbb{Z}, +)$, which is a cyclic group of order 6. The elements of this group are $[0], [1], [2], [3], [4], [5]$. The identity element is $[0]$. The order of the element $[1]$ is 6, since $[1]^6 = [6] = [0]$ and there is no smaller positive integer k such that $[1]^k = [0]$. Similarly, the order of the element $[2]$ is 3, since $[2]^3 = [6] = [0]$ and there is no smaller positive integer k such that $[2]^k = [0]$. The order of the element $[3]$ is 2, since $[3]^2 = [6] = [0]$ and there is no smaller positive integer k such that $[3]^k = [0]$. The order of the element $[4]$ is 3, since $[4]^3 = [12] = [0]$ and there is no smaller positive integer k such that $[4]^k = [0]$. The order of the element $[5]$ is 6, since $[5]^6 = [30] = [0]$ and there is no smaller positive integer k such that $[5]^k = [0]$.
Example 1.3.0.2 **-**
Consider the group $(\mathbb{Z}, +)$, which is an infinite cyclic group. The elements of this group are all integers. The identity element is 0. The order of any non-zero element $n \in \mathbb{Z}$ is infinite, since there is no positive

integer k such that $n^k = 0$. The order of the element 0 is 1, since $0^1 = 0$ and there is no smaller positive integer k such that $0^k = 0$.

1.4 Dihedral Groups

The symmetries of any geometric object form a group under the operation of composition of symmetries. For example, the symmetries of a regular polygon with n sides form a group called the dihedral group of order $2n$, denoted by D_n. The elements of D_n consist of n rotations and n reflections. The group operation is the composition of symmetries, which is associative. The identity element is the identity transformation (which leaves the polygon unchanged), and each element has an inverse (the symmetry that undoes it). The dihedral group D_n is non-abelian for $n \geq 3$, since the composition of a rotation and a reflection does not commute. Let us consider an n sided polygon, and denote the rotation by r and the reflection by f. We want to identify the elements of D_n and the relations between them. We have the following relations -

- $r^n = e$, since applying the rotation n times results in the identity transformation.
- $f^2 = e$, since applying the reflection twice results in the identity transformation.
- $frf^{-1} = r^{-1}$, since conjugating a rotation by a reflection results in the inverse of the rotation.

From the above relations, we can deduce that the elements of D_n can be expressed in the form r^k for $k = 0, 1, \ldots, n-1$ (representing the rotations) and fr^k for $k = 0, 1, \ldots, n-1$ (representing the reflections). Thus, the dihedral group D_n has $2n$ elements, and it is generated by the rotation r and the reflection f. A formal definition of the dihedral group D_n can be given as follows -

Definition 1.4.0.1

Dihedral Group
The dihedral group of order $2n$, denoted by D_n, is the group defined by the presentation:

$$D_n = \langle r, f \mid r^n = e, f^2 = e, frf^{-1} = r^{-1} \rangle$$

where r represents a rotation and f represents a reflection. The elements of D_n can be expressed as r^k for $k = 0, 1, \ldots, n-1$ (representing the rotations) and fr^k for $k = 0, 1, \ldots, n-1$ (representing the reflections). The group operation is the composition of symmetries.

1.5 Symmetric Groups

Definition 1.5.0.1

Permutation
A permutation of a set S is a bijection from S to itself. In other words, a permutation is a rearrangement of the elements of S. For example, if $S = \{1, 2, 3\}$, then one possible permutation is the function that maps 1 to 2, 2 to 3, and 3 to 1. This can be represented in cycle notation as (1 2 3).

We note that the set of all permutations of a set S forms a group under the operation of composition of functions. The identity element in this group is the identity permutation (which maps each element to itself), and each element has an inverse (the permutation that undoes it). The group of all permutations of a set S is called the symmetric group on S. If S has n elements, then the symmetric group on S is denoted by S_n and has $n!$ elements, since there are $n!$ ways to rearrange n distinct objects.

Definition 1.5.0.2

Symmetric Group
The symmetric group on n elements, denoted by S_n, is the group of all permutations of n distinct objects. The elements of S_n are the bijections from the set $\{1, 2, \ldots, n\}$ to itself, and the group

operation is the composition of these bijections, defined as folows:

$$\star : S_n \times S_n \to S_n$$

$$(\sigma, \tau) \mapsto \sigma \cdot \tau$$

where $\sigma \cdot \tau$ is the permutation obtained by first applying τ and then applying σ. The identity element in S_n is the identity permutation, which maps each element to itself, and the inverse of a permutation is the permutation that undoes it.

We denote each element of S_n as follows -

$$\begin{pmatrix} 1\ 2\ \ldots\ n \\ \sigma(1)\ \sigma(2)\ \ldots\ \sigma(n) \end{pmatrix}$$

where σ is a permutation of the set $\{1, 2, \ldots, n\}$. For example, the element of S_3 that maps 1 to 2, 2 to 3, and 3 to 1 can be represented as:

$$\begin{pmatrix} 1\ 2\ 3 \\ 2\ 3\ 1 \end{pmatrix}$$

We now define a cycle in a symmetric group.

Definition 1.5.0.3

Cycle
A k-cycle in a symmetric group S_n is a permutation that maps a subset of $k \leq n$ distinct elements to each other in a cyclic manner, while leaving the remaining elements fixed. For example, the 3-cycle (1 2 3) in S_3 maps 1 to 2, 2 to 3, and 3 to 1, while leaving any other elements (if they exist) fixed.

A cycle can be represented in cycle notation, where the elements that are permuted are enclosed in parentheses. For example, the 3-cycle (1 2 3) can be represented as:

$$\begin{pmatrix} 1\ 2\ 3 \\ 2\ 3\ 1 \end{pmatrix}$$

Definition 1.5.0.4

Disjoint Cycles
Two cycles in a symmetric group S_n are said to be disjoint if the sets they permute are disjoint. For example, the cycles (1 2) and (3 4) in S_4 are disjoint, since they permute different sets of elements. On the other hand, the cycles (1 2) and (2 3) in S_3 are not disjoint, since they both permute the element 2.

We now state the following results -

Theorem 1.2 (Disjoint Cycles Commute). *If c_1 and c_2 are disjoint cycles in a symmetric group S_n, then they commute with each other. In other words, if c_1 and c_2 are disjoint cycles, then $c_1c_2 = c_2c_1$.*

Proof. Let c_1 and c_2 be disjoint cycles in S_n. Since c_1 and c_2 permute different sets of elements, we can write $c_1 = (a_1\ a_2\ \ldots\ a_k)$ and $c_2 = (b_1\ b_2\ \ldots\ b_m)$, where a_i and b_j are distinct elements of the set $\{1, 2, \ldots, n\}$.

Now, we want to show that $c_1c_2 = c_2c_1$. Let $x \in \{1, 2, \ldots, n\}$. We have three cases to consider:

- If $x = a_i$ for some i, then $c_1(x) = a_{i+1}$ (where we take indices modulo k) and $c_2(x) = x$. Therefore, we have:

$$(c_1c_2)(x) = c_1(c_2(x)) = c_1(x) = a_{i+1}$$

and

$$(c_2c_1)(x) = c_2(c_1(x)) = c_2(a_{i+1}) = a_{i+1}$$

Thus, $(c_1c_2)(x) = (c_2c_1)(x)$.

- If $x = b_j$ for some j, then $c_1(x) = x$ and $c_2(x) = b_{j+1}$ (where we take indices modulo m). Therefore, we have:
$$(c_1c_2)(x) = c_1(c_2(x)) = c_1(b_{j+1}) = b_{j+1}$$
and
$$(c_2c_1)(x) = c_2(c_1(x)) = c_2(x) = b_{j+1}$$
Thus, $(c_1c_2)(x) = (c_2c_1)(x)$.

- If x is not permuted by either c_1 or c_2, then $c_1(x) = x$ and $c_2(x) = x$. Therefore, we have:
$$(c_1c_2)(x) = c_1(c_2(x)) = c_1(x) = x$$
and
$$(c_2c_1)(x) = c_2(c_1(x)) = c_2(x) = x$$
Thus, $(c_1c_2)(x) = (c_2c_1)(x)$.

Since $(c_1c_2)(x) = (c_2c_1)(x)$ for all $x \in \{1, 2, \ldots, n\}$, we conclude that $c_1c_2 = c_2c_1$. Therefore, disjoint cycles commute with each other. □

Theorem 1.3 (Decomposition into Disjoint Cycles)**.** *Every permutation in a symmetric group S_n can be uniquely expressed as a product of disjoint cycles. In other words, for any permutation $\sigma \in S_n$, there exist disjoint cycles $c_1, c_2, \ldots, c_k$ such that $\sigma = c_1c_2 \cdots c_k$, and this decomposition is unique up to the order of the cycles.*

Proof. Let $\sigma \in S_n$ be a permutation. We will construct the disjoint cycles that represent σ. Start with an element i_1 in the set $\{1, 2, \ldots, n\}$. Apply σ to i_1 to get $i_2 = \sigma(i_1)$. Continue applying σ to the resulting elements until we return to i_1. This process will give us a cycle that permutes the elements $i_1, i_2, \ldots, i_k$, where $k \leq n$ is the length of the cycle.

Next, choose an element j_1 that has not been permuted by the first cycle. Repeat the same process to find another cycle that permutes a different set of elements. Continue this process until all elements have been permuted by some cycle. Since each element can only be permuted by one cycle, the cycles we obtain are disjoint. Now, we show we can express σ as the product of these disjoint cycles. Let $c_1, c_2, \ldots, c_m$ be the disjoint cycles obtained from the process above. Since the cycles are disjoint, they commute with each other. Therefore, we can write:
$$\sigma = c_1c_2 \cdots c_m$$
□

Theorem 1.4 (Order of a k-cycle)**.** *The order of a k-cycle in a symmetric group S_n is k. In other words, if σ is a k-cycle, then $\sigma^k = e$, where e is the identity permutation, and there is no smaller positive integer $m < k$ such that $\sigma^m = e$.*

Proof. Let $\sigma = (a_1\ a_2\ \ldots\ a_k)$ be a k-cycle in S_n. We want to show that $\sigma^k = e$ and that there is no smaller positive integer $m < k$ such that $\sigma^m = e$.

First, we compute σ^2:
$$\sigma^2(a_1) = \sigma(a_2) = a_3, \quad \sigma^2(a_2) = \sigma(a_3) = a_4, \quad \ldots, \quad \sigma^2(a_{k-1}) = \sigma(a_k) = a_1, \quad \sigma^2(a_k) = \sigma(a_1) = a_2$$
Thus, $\sigma^2 = (a_1\ a_3\ a_5\ \ldots)(a_2\ a_4\ a_6\ \ldots)$. Continuing this process, we find that:
$$\sigma^3(a_1) = a_4, \quad \sigma^3(a_2) = a_5, \quad \ldots$$
After applying $k - 1$ times, we have:
$$\sigma^{k-1}(a_1) = a_k, \quad \sigma^{k-1}(a_2) = a_1, \quad \ldots$$
Finally, we compute $\sigma^k(a_i) = a_i$ for all $i = 1, 2, ..., k$, which shows that $\sigma^k = e$.

Now, suppose there exists a positive integer $m < k$ such that $\sigma^m = e$. This would imply that applying the cycle m times results in the identity permutation. However, since the cycle permutes k distinct elements in a cyclic manner, it would take exactly k applications of the cycle to return to the original configuration. Therefore, there cannot be any smaller positive integer $m < k$ such that $\sigma^m = e$. □

Theorem 1.5 (Order of a Product of Disjoint Cycles)**.** *The order of a product of disjoint cycles in a symmetric group S_n is the least common multiple (LCM) of the lengths of the cycles. In other words, if $\sigma = c_1 c_2 \cdots c_m$ is a product of disjoint cycles, where c_i is a cycle of length k_i, then the order of σ is given by:*

$$ord(\sigma) = lcm(k_1, k_2, \ldots, k_m)$$

Proof. Let $\sigma = c_1 c_2 \cdots c_m$ be a product of disjoint cycles, where c_i is a cycle of length k_i. Since the cycles are disjoint, they commute with each other. Therefore, we can write:

$$\sigma^n = (c_1 c_2 \cdots c_m)^n = c_1^n c_2^n \cdots c_m^n$$

We want to find the smallest positive integer n such that $\sigma^n = e$. This is equivalent to finding the smallest positive integer n such that $c_i^n = e$ for all $i = 1, 2, ..., m$.

Since c_i is a cycle of length k_i, we know from the previous theorem that $c_i^{k_i} = e$ and there is no smaller positive integer $m < k_i$ such that $c_i^m = e$. Therefore, for each cycle c_i, the order is k_i.

To ensure that $c_i^n = e$ for all $i = 1, 2, ..., m$, we need to find the smallest positive integer n that is a multiple of each k_i. This is precisely the definition of the least common multiple (LCM). Thus, we have:

$$n = \text{lcm}(k_1, k_2, ..., k_m)$$

Therefore, the order of the product of disjoint cycles σ is given by:

$$\text{ord}(\sigma) = n = \text{lcm}(k_1, k_2, ..., k_m)$$

□

Theorem 1.6 (Number of k-cycles in S_n)**.** *The number of k-cycles in the symmetric group S_n is given by the formula:*

$$(k-1)! \cdot \binom{n}{k}$$

where $\binom{n}{k}$ is the binomial coefficient representing the number of ways to choose k elements from a set of n elements, and $(k-1)!$ accounts for the number of distinct cycles that can be formed with those k elements.

Proof. Choose k distinct elements from the set $\{1, 2, \ldots, n\}$ to form the cycle. The number of ways to choose k elements from n is given by the binomial coefficient:

$$\binom{n}{k}$$

Once we have chosen the k elements, we can arrange them in a cycle. The number of ways to arrange k distinct elements in a cycle is given by $(k-1)!$, since we can fix one element and permute the remaining $k-1$ elements in any order. Therefore, the total number of k-cycles in S_n is given by:

$$(k-1)! \cdot \binom{n}{k}$$

□

We now define a two cycle in a symmetric group.

Definition 1.5.0.5

Two Cycle
A two cycle (also known as a transposition) in a symmetric group S_n is a cycle that permutes exactly two elements while leaving the remaining elements fixed. For example, the two cycle (1 2) in S_3 maps 1 to 2, 2 to 1, and leaves 3 fixed.

Note 1.5.0.1

We note that for all transpositions σ, we have $\sigma^2 = e$, since applying the transposition twice results in the identity permutation. It immediately follows that $\sigma^{-1} = \sigma$

Definition 1.5.0.6

Parity of a Permutation
The parity of a permutation $\sigma \in S_n$ is defined as follows:

- If σ can be expressed as a product of an even number of transpositions, then σ is called an even permutation, and its parity is defined to be 0.
- If σ can be expressed as a product of an odd number of transpositions, then σ is called an odd permutation, and its parity is defined to be 1.

Theorem 1.7 (Any cycle can be expressed as a product of transpositions)**.** *Every cycle in a symmetric group S_n can be expressed as a product of transpositions. In particular, a k-cycle can be expressed as a product of $k-1$ transpositions.*

Proof. Let $\sigma = (a_1\ a_2\ \ldots\ a_k)$ be a k-cycle in S_n. We can express σ as a product of transpositions as follows:

$$\sigma = (a_1\ a_k)(a_1\ a_{k-1})\cdots(a_1\ a_2)$$

To verify that this product of transpositions is equal to the original cycle σ, we can apply both sides to an arbitrary element $x \in \{1, 2, \ldots, n\}$ and check that they yield the same result. We have three cases to consider:

- If $x = a_1$, then the right-hand side maps a_1 to a_2, which is the same as the left-hand side.
- If $x = a_i$ for some $i = 2, 3, \ldots, k$, then the right-hand side maps a_i to a_{i+1} (where we take indices modulo k), which is the same as the left-hand side.
- If x is not one of the elements in the cycle, then both sides map x to itself.
- Thus, we have shown that $\sigma = (a_1\ a_k)(a_1\ a_{k-1})\cdots(a_1\ a_2)$, which is a product of $k-1$ transpositions. Therefore, every cycle can be expressed as a product of transpositions.

□

Corollary 1.7.1:

Every permutation in a symmetric group S_n can be expressed as a product of transpositions. In particular, if a permutation σ can be expressed as a product of disjoint cycles, then we can express each cycle as a product of transpositions, and thus express σ as a product of transpositions.

Proof. Let $\sigma \in S_n$ be a permutation. From the theorem on decomposition into disjoint cycles, we can express σ as a product of disjoint cycles:

$$\sigma = c_1 c_2 \cdots c_m$$

where each c_i is a cycle. From the previous theorem, we know that each cycle c_i can be expressed as a product of transpositions. Therefore, we can write:

$$\sigma = (t_{i1} t_{i2} \cdots t_{i l_i})_{i=1}^{m}$$

where each t_{ij} is a transposition. This shows that σ can be expressed as a product of transpositions. □

Theorem 1.8 (Number of odd and even permutations in S_n)**.** *The number of even permutations in the symmetric group S_n is equal to the number of odd permutations, and both are equal to $\frac{n!}{2}$. In other words, there are $\frac{n!}{2}$ even permutations and $\frac{n!}{2}$ odd permutations in S_n.*

Proof. We can establish a bijection between the set of even permutations and the set of odd permutations in S_n by fixing a transposition t and defining a function $f : S_{n_e} \to S_{n_0}$ as follows:

$$f(\sigma) = t \cdot \sigma$$

for any even permutation $\sigma \in S_{n_e}$.

We need to show that f is a bijection. First, we show that f is injective. Suppose $f(\sigma_1) = f(\sigma_2)$ for some $\sigma_1, \sigma_2 \in S_n$. This means that:

$$t \cdot \sigma_1 = t \cdot \sigma_2$$

Multiplying both sides by $t^{-1} = t$, we get:

$$t^{-1} \cdot (t \cdot \sigma_1) = t^{-1} \cdot (t \cdot \sigma_2)$$

which simplifies to:

$$\sigma_1 = \sigma_2$$

Thus, f is injective.

Next, we show that f is surjective. Let $y \in S_{n_o}$ be any permutation. We want to find a permutation $x \in S_{n_e}$ such that $f(x) = y$. We can choose $x = t^{-1}y = ty$, since $t^{-1} = t$. Then we have:

$$f(x) = f(ty) = t(ty) = (tt)y = ey = y$$

Thus, for every permutation $y \in S_{n_o}$, there exists a permutation $x = ty$ such that $f(x) = y$. Therefore, f is surjective.

Since f is both injective and surjective, it is a bijection. This means that the number of even permutations in S_n is equal to the number of odd permutations in S_n. Since the total number of permutations in S_n is $n!$, we have:

$$\text{Number of even permutations} + \text{Number of odd permutations} = n!$$

Let E be the number of even permutations and O be the number of odd permutations. We have:

$$E = O = \frac{n!}{2}$$

Hence proved. □

Theorem 1.9 (S_{n_e} is a subgroup of S_n)**.** *The set of even permutations in the symmetric group S_n, denoted by S_{n_e}, forms a subgroup of S_n. In other words, S_{n_e} is closed under the group operation (composition of permutations), contains the identity element, and contains the inverse of each of its elements.*

Proof. To show that S_{n_e} is a subgroup of S_n, we need to verify the following three properties:

- **Closure:** If $\sigma_1, \sigma_2 \in S_{n_e}$, then $\sigma_1 \cdot \sigma_2 \in S_{n_e}$.
- **Identity:** The identity permutation e is in S_{n_e}.
- **Inverses:** If $\sigma \in S_{n_e}$, then the inverse permutation σ^{-1} is also in S_{n_e}.

Closure: Let $\sigma_1, \sigma_2 \in S_{n_e}$ be two even permutations. By definition, σ_1 can be expressed as a product of an even number of transpositions, and σ_2 can also be expressed as a product of an even number of transpositions. When we compose σ_1 and σ_2, the resulting permutation $\sigma_1 \cdot \sigma_2$ can be expressed as a product of the transpositions from σ_1 and the transpositions from σ_2. Since both σ_1 and σ_2 are even, the total number of transpositions in the composition $\sigma_1 \cdot \sigma_2$ is also even. Therefore, $\sigma_1 \cdot \sigma_2 \in S_{n_e}$.
Identity: The identity permutation e can be expressed as a product of zero transpositions, which is an even number. Therefore, $e \in S_{n_e}$.
Inverses: Let $\sigma \in S_{n_e}$ be an even permutation. Since σ can be expressed as a product of an even number of transpositions, we can write:

$$\sigma = t_1 t_2 \cdots t_{2k}$$

where t_i are transpositions and $2k$ is an even number. The inverse of σ can be expressed as:

$$\sigma^{-1} = t_{2k}^{-1} t_{2k-1}^{-1} \cdots t_1^{-1}$$

Since each transposition is its own inverse (i.e., $t_i^{-1} = t_i$), we have:

$$\sigma^{-1} = t_{2k}t_{2k-1}\cdots t_1$$

This is also a product of an even number of transpositions, so $\sigma^{-1} \in S_{n_e}$.

Since S_{n_e} satisfies closure, contains the identity element, and contains the inverses of its elements, we conclude that S_{n_e} is a subgroup of S_n. □

Theorem 1.10 (Transpositions as generators of S_n)**.** *The symmetric group S_n is generated by the set of transpositions given by $(1\ 2), (2\ 3), \ldots, (n-1\ n)$. In other words, every permutation in S_n can be expressed as a product of these transpositions.*

Proof. We will show that any transposition can be expressed as a product of the transpositions $(1\,2), (2\,3), \ldots, (n-1\ n)$. Since every permutation can be expressed as a product of transpositions, it will follow that every permutation can be expressed as a product of the given transpositions.

Let $(i\ j)$ be a transposition where $1 \le i < j \le n$. We want to express $(i\ j)$ as a product of the transpositions $(1\ 2), (2\ 3), \ldots, (n-1\ n)$.

We can achieve this by first moving i to position 1 and j to position 2 using the given transpositions, then applying the transposition $(1\ 2)$, and finally moving the elements back to their original positions. The sequence of transpositions can be written as follows:

$$(i\ j) = (1\ i)(1\ j)(1\ i)$$

This expression shows that we can express any transposition $(i\ j)$ as a product of the transpositions $(1\ 2), (2\ 3), \ldots, (n-1\ n)$.

Since every permutation in S_n can be expressed as a product of transpositions, and we have shown that every transposition can be expressed as a product of the given transpositions, it follows that every permutation in S_n can be expressed as a product of the transpositions $(1\,2), (2\,3), \ldots, (n-1\,n)$. Therefore, these transpositions generate the symmetric group S_n. □

Note 1.5.0.2

We note that a set of $n-2$ or less transpositions cannot generate S_n. For example, the set of transpositions $\{(1\ 2), (2\ 3), \ldots, (n-2\ n-1)\}$ cannot generate S_n.
We also note that there are n^{n-2} different ways to generate S_n using $n-1$ transpositions.

1.5.1 Cycle Type

Definition 1.5.1.1

Cycle Type
The cycle type of a permutation $\sigma \in S_n$ is a way to classify permutations based on the lengths of the cycles in their disjoint cycle decomposition. The cycle type is often represented as a partition of n that indicates how many cycles of each length are present in the decomposition. For example, if a permutation has two 1-cycles (fixed points), one 2-cycle, and one 3-cycle, its cycle type can be represented as $(1^2, 2^1, 3^1)$.

We note the following facts:

- $\sum_{k=1}^{n} k \cdot m_k = n$, where m_k is the number of cycles of length k in the cycle decomposition of the permutation.
- $k_1 + k_2 + \ldots + k_m$ forms a partition of n, where k_i is the length of the i-th cycle arranged in ascending order in the cycle decomposition of the permutation.

Theorem 1.11 (Number of permutations with a cycle type)**.** *The number of permutations in S_n with a given cycle type $(1^{m_1}, 2^{m_2}, \ldots, n^{m_n})$ is given by the formula:*

$$\frac{n!}{\prod_{k=1}^{n}(k^{m_k} \cdot m_k!)}$$

where m_k is the number of cycles of length k in the cycle decomposition of the permutation.

Proof. The number of possible arrangements of n symbols is given by $n!$. However, we need to account for the indistinguishability of cycles of the same length and the indistinguishability of elements within each cycle. For each cycle of length k, there are k ways to arrange the elements within the cycle, and for m_k cycles of length k, there are $m_k!$ ways to arrange the cycles themselves. Therefore, we need to divide $n!$ by the product of k^{m_k} (to account for the arrangements within each cycle) and $m_k!$ (to account for the arrangements of the cycles). We need to consider the overcounting for all k from 1 to n, which gives us the formula:

$$\frac{n!}{\prod_{k=1}^{n}(k^{m_k} \cdot m_k!)}$$

Hence proved. □

1.6 Lagrange's Theorem

Definition 1.6.0.1

Cosets
Let G be a group and H be a subgroup of G. For any element $g \in G$, the left coset of H in G with respect to g is defined as the set:

$$gH = \{gh : h \in H\}$$

Similarly, the right coset of H in G with respect to g is defined as the set:

$$Hg = \{hg : h \in H\}$$

It is easy to see that if we have $g \in H$, then the left coset gH is just H itself, and the right coset Hg is also just H. However, if $g \notin H$, then the left coset gH and the right coset Hg are different from H. In fact, they are disjoint from H and from each other.
Further, we should also note that the left coset gH and the right coset Hg may not be the same. For example, if we take $G = S_3$ and $H = \{e, (1\ 2)\}$, then the left coset of H with respect to $(1\ 3)$ is $(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\}$, while the right coset of H with respect to $(1\ 3)$ is $H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\}$. Thus, in this case, the left coset and the right coset are different. However, for abelian groups, the left coset and the right coset are always the same, since the group operation is commutative. In general, for non-abelian groups, the left coset and the right coset may be different. Now, suppose, we have a finite group G and a subgroup H of G and let $g \in G$ be some fixed element, $h \in H$. We can state the following -

- If $gh = gk$ for some $k \in H$, then $h = k$. This means that the left coset gH has the same number of elements as H.
- Either $gH = g'H$ or $gH \cap g'H = \emptyset$ for any $g, g' \in G$. This means that the left cosets of H in G are either identical or disjoint. The same is true for the right cosets.

We prove the second point as follows -

Proof. Let $g, g' \in G$ be two elements such that $gH \cap g'H \neq \emptyset$. This means that there exists an element $x \in G$ such that $x \in gH$ and $x \in g'H$. Therefore, we can write:

$$x = gh_1 = g'h_2$$

for some $h_1, h_2 \in H$. Rearranging this equation gives us:

$$g^{-1}g' = h_1h_2^{-1}$$

Since $h_1h_2^{-1} \in H$, we have $g^{-1}g' \in H$. This implies that $g' = gh$ for some $h \in H$. Therefore, we can write:

$$g'H = (gh)H = g(Hh) = gH$$

Thus, if the left cosets gH and $g'H$ are not disjoint, they must be identical. The same argument applies to the right cosets. Hence proved. □

we can clearly see that the left cosets of H in G partition the group G into disjoint subsets, and the same is true for the right cosets. Moreover, each left coset has the same number of elements as H, and each right coset also has the same number of elements as H. Let us take an example to illustrate this fact -

Example 1.6.0.1 **-**

Consider the finite group $G = S_3$ and the subgroup $H = \{e, (1\ 2)\}$. The left cosets of H in G are:

$$eH = \{e, (1\ 2)\}$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\}$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\}$$

Taking the union of these left cosets, we have:

$$eH \cup (1\ 3)H \cup (2\ 3)H = \{e, (1\ 2), (1\ 3), (1\ 2\ 3), (2\ 3), (1\ 3\ 2)\} = S_3$$

Thus, the left cosets of H in G partition the group G into disjoint subsets, and each left coset has the same number of elements as H. Similarly, we can find the right cosets of H in G and verify that they also partition G into disjoint subsets with the same number of elements as H.

We prove that the left cosets of H in G partition the group G into disjoint subsets as follows -

Proof. Let G be a finite group and H be a subgroup of G. We want to show that the left cosets of H in G partition the group G into disjoint subsets.

First, we show that every element of G belongs to at least one left coset of H. Let $g \in G$ be an arbitrary element. Then, the left coset of H with respect to g is given by:

$$gH = \{gh : h \in H\}$$

Since $e \in H$, we have $g = ge \in gH$. Therefore, every element of G belongs to at least one left coset of H.

Next, we show that the left cosets of H are either identical or disjoint. Let $g, g' \in G$ be two elements such that $gH \cap g'H \neq \emptyset$. This means that there exists an element $x \in G$ such that $x \in gH$ and $x \in g'H$. Therefore, we can write:

$$x = gh_1 = g'h_2$$

for some $h_1, h_2 \in H$. Rearranging this equation gives us:

$$g^{-1}g' = h_1h_2^{-1}$$

Since $h_1h_2^{-1} \in H$, we have $g^{-1}g' \in H$. This implies that $g' = gh$ for some $h \in H$. Therefore, we can write:

$$g'H = (gh)H = g(Hh) = gH$$

Thus, if the left cosets gH and $g'H$ are not disjoint, they must be identical.

Since every element of G belongs to at least one left coset of H, and any two left cosets are either identical or disjoint, it follows that the left cosets of H in G partition the group G into disjoint subsets. The same argument applies to the right cosets. Hence proved. □

Now, we are equipped to break a finite group G into cosets of any subgroup H of G. Since all cosets of H have the same number of elements as H and the same number of elements pairwise, we can conclude that the order of G is a multiple of the order of H. This leads us to the statement of Lagrange's theorem.

Theorem 1.12 (Lagrange's Theorem). *Let G be a finite group and H be a subgroup of G. Then the order of H divides the order of G.*

Proof. Let G be a finite group and H be a subgroup of G. We will show that the order of H divides the order of G. Let $|G| = n$ denote the order of G and $|H| = m$ denote the order of H.

Trivial case: If $H = \{e\}$ is the trivial subgroup, then $|H| = 1$ divides $|G|$ for any finite group G. Thus, the theorem holds in this case. Further if $H = G$, then $|H| = |G|$ and the theorem holds.

Non-trivial case: Assume H is a non-trivial proper subgroup of G. We will use the concept of left cosets to partition G into disjoint subsets. Pick $g_1 \in G, g_1 \notin H$. Then we can form the left coset g_1H. Since $g_1 \notin H$, the left coset g_1H is disjoint from H. Next, take $g_2 \in G \setminus \{H \cup g_1H\}$ and form the left coset g_2H. This left coset is also disjoint from both H and g_1H. We can continue this process until we have partitioned G into disjoint left cosets of H:

$$G = H \cup g_1H \cup g_2H \cup \ldots \cup g_kH$$

where k is the number of left cosets of H in G. Since each left coset has the same number of elements as H, we have:

$$|G| = |H| + |g_1H| + |g_2H| + \ldots + |g_kH| = (k+1)|H|$$

This shows that the order of G is a multiple of the order of H. Therefore, we can conclude that the order of H divides the order of G. Hence proved. □

This is a powerful result which says that subgroups cannot have arbitrary orders; they must divide the order of the group. For example, if G is a group of order 12, then any subgroup H of G must have an order that divides 12, which means H can only have orders 1, 2, 3, 4, 6, or 12. This theorem has many applications in group theory and helps us understand the structure of groups and their subgroups.

Note 1.6.0.1

We must note that the converse is not true, i.e., if d divides the order of G, it does not necessarily mean that there is a subgroup of G with order d. For example, the group A_4 (the alternating group on 4 elements) has order 12, but it does not have a subgroup of order 6, even though 6 divides 12.

Corollary 1.12.1:

In a finite group G, the order of any element $g \in G$ divides the order of G.

Proof. Let G be a finite group and $g \in G$ be an arbitrary element. Consider the subgroup generated by g, denoted by $\langle g \rangle$. The order of the subgroup $\langle g \rangle$ is equal to the order of the element g, which we denote by $o(g)$. Since $\langle g \rangle$ is a subgroup of G, we can apply Lagrange's theorem, which states that the order of any subgroup divides the order of the group. Therefore, we have:

$$o(g) = |\langle g \rangle| \text{ divides } |G|$$

Hence proved. □

Corollary 1.12.2:

If G is a finite group of prime order p, then G is cyclic and thus isomorphic to the additive group of integers modulo p, denoted by $\mathbb{Z}_p$.

Proof. Let G be a finite group of prime order p. Since p is prime, the only divisors of p are 1 and p. By Lagrange's theorem, the order of any subgroup of G must divide p. Therefore, the only subgroups of G are the trivial subgroup $\{e\}$ and the group G itself.

Now, let $g \in G$ be any non-identity element. The subgroup generated by g, denoted by $\langle g \rangle$, must have an order that divides p. Since g is not the identity element, the order of g cannot be 1. Therefore, the order of g must be p. This means that g generates the entire group G, and thus $G = \langle g \rangle$.

Since G is generated by a single element, it is cyclic. Moreover, a cyclic group of prime order is isomorphic to the additive group of integers modulo that prime. Therefore, we have:

$$G \cong \mathbb{Z}_p$$

Hence proved. □

1.6.1 Classifying Groups of Order 4

Let $(G, \cdot)$ be a group of order 4. We want to classify all possible groups of order 4 up to isomorphism. We can use Lagrange's theorem to determine the possible orders of elements in G. Since G has order 4, the possible orders of elements are 1, 2, or 4.

- If G has an element of order 4, then G is cyclic and isomorphic to $\mathbb{Z}_4$.

- If G has no element of order 4, but has an element of order 2, then G must have two distinct elements of order 2 (since the identity element has order 1). In this case, G is isomorphic to the Klein four-group V_4, which can be represented as:

$$V_4 = \{e, a, b, ab\}$$

where $a^2 = b^2 = e$ and $ab = ba$.

Thus, there are exactly two groups of order 4 up to isomorphism: the cyclic group $\mathbb{Z}_4$ and the Klein four-group V_4. We can note that all groups of order 4 are abelian, since both $\mathbb{Z}_4$ and V_4 are abelian groups. Therefore, there are no non-abelian groups of order 4. This classification of groups of order 4 is a direct consequence of Lagrange's theorem and the properties of group elements.

Exercise 1.8: Prove the following statements:

1. Every group of order less than 6 is abelian.
2. If $H \leq G$ such that $|G| = 6$, then H is cyclic.
3. Let $n > 2$ be a prime number. Suppose we have a non commutative group G of order $2n$, then G has a subgroup of order n.
4. Suppose G is a finite group, then for every element $a \in G$, $a^{|G|} = e$.
5. Suppose G is a finite group and we have k such that $\gcd(k, |G|) = 1$, then the map $f : G \to G$ defined by $f(g) = g^k$ is a bijection.

Solution: We prove each part seperately.

1. Let G be a group of order less than 6. The possible orders of G are 1, 2, 3, 4, or 5. If G has order 1, it is the trivial group and is abelian. If G has order 2 or 3 or 5, it is cyclic (since any group of prime order is cyclic) and thus abelian. If G has order 4, it can either be isomorphic to $\mathbb{Z}_4$ or V_4, both of which are abelian. Therefore, every group of order less than 6 is abelian.

2. Let G be a group of order 6 and H be a subgroup of G. The possible orders of H are 1, 2, 3. If H has order 1, it is the trivial subgroup and is cyclic. If H has order 2 or 3, it is cyclic (since any group of prime order is cyclic).

3. Let G be a non-commutative group of order $2n$ where $n > 2$ is a prime number. By Lagrange's theorem, the possible orders of subgroups of G are 1, 2, n. Suppose it does not have a subgroup of order n. Then the only subgroups are the trivial subgroup and subgroups of order 2. Now, this would imply that all elements of G other than identity have order 2. Consider two such elements a and b. Since G is non-commutative, we must have $ab \neq ba$. However, since both a and b have order 2, we have:
$$(ab)^2 = abab = a(ba)b = a(ab)b = (aa)(bb) = e$$
This is a contradiction since this means that the group operation is commutative. Therefore, G must have a subgroup of order n.

4. Suppose G is a finite group and let $a \in G$. Let $o(a)$ denote the order of the element a. By Lagrange's theorem, the order of any element divides the order of the group. Therefore, we can write:
$$o(a) \text{ divides } |G|$$
This means that there exists an integer k such that $o(a) \cdot k = |G|$. Therefore, we have:
$$a^{|G|} = a^{o(a)\cdot k} = (a^{o(a)})^k = e^k = e$$
Hence proved.

5. Suppose G is a finite group and let k be an integer such that $\gcd(k, |G|) = 1$. We want to show that the map $f : G \to G$ defined by $f(g) = g^k$ is a bijection. First, we show that f is injective. Suppose $f(g_1) = f(g_2)$ for some $g_1, g_2 \in G$. This means that:
$$g_1^k = g_2^k$$

Rearranging this gives us:

$$g_1^k g_2^{-k} = e$$

This implies that $(g_1 g_2^{-1})^k = e$. Since $\gcd(k, |G|) = 1$, the order of $g_1 g_2^{-1}$ must divide k and $|G|$. Therefore, $g_1 g_2^{-1} = e$, which means $g_1 = g_2$. Thus, f is injective. Next, we show that f is surjective. Let $y \in G$ be an arbitrary element. We want to find an element $x \in G$ such that $f(x) = y$. Since $\gcd(k, |G|) = 1$, there exist integers m and n such that $mk + n|G| = 1$. Recall note 1.2.2.4. Since we have $k, |G|$ are coprime, the identity demonstrated still holds. Therefore, we can write:

$$y = y^{mk+n|G|} = y^{mk} \cdot y^{n|G|} = (y^m)^k \cdot e = (y^m)^k$$

Let $x = y^m$. Then we have:

$$f(x) = f(y^m) = (y^m)^k = y$$

Thus, for every element $y \in G$, there exists an element $x = y^m \in G$ such that $f(x) = y$. Therefore, f is surjective.

1.7 Group Homomorphisms

Let us compare two groups for structural similarity. Consider the group of integers under addition, denoted by $(\mathbb{Z}, +)$, and the group of integers modulo 2 under addition, denoted by $(\mathbb{Z}_2, +)$. We can partition the set of integers into two subsets: the even integers and the odd integers. The even integers form a subgroup of $\mathbb{Z}$ isomorphic to $\mathbb{Z}$ itself, while the odd integers do not form a subgroup. Consider adding elements of these sets in different ways -

$(\mathbb{Z}, +)$	$(\mathbb{Z}_2, +)$
even + even = even	$0 + 0 \equiv 0 \mod 2$
odd + odd = even	$1 + 1 \equiv 0 \mod 2$
even + odd = odd	$0 + 1 \equiv 1 \mod 2$
odd + even = odd	$1 + 0 \equiv 1 \mod 2$

From this comparison, we can see that the structure of the group $(\mathbb{Z}, +)$ is similar to the structure of the group $(\mathbb{Z}_2, +)$ in terms of how elements combine under the group operation. We can define a function $f : \mathbb{Z} \to \mathbb{Z}_2$ by:

$$f(n) = \begin{cases} 0 & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd} \end{cases}$$

This function preserves the group operation, since for any $m, n \in \mathbb{Z}$, we have:

$$f(m+n) = \begin{cases} 0 & \text{if } m+n \text{ is even} \\ 1 & \text{if } m+n \text{ is odd} \end{cases}$$

and

$$f(m) + f(n) = \begin{cases} 0 & \text{if } m \text{ and } n \text{ are both even or both odd} \\ 1 & \text{if } m \text{ and } n \text{ have different parity} \end{cases}$$

Let us now consider more examples to illustrate this more concretely.
Consider the group Z_4 under addition modulo 4 and the group of complex numbers of the form $\{1, i, -1, -i\}$ under multiplication. Let us construct the cayley tables for both groups -

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$\cdot$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

From the cayley tables, we can see that the structure of the group Z_4 under addition modulo 4 is similar to the structure of the group of complex numbers $\{1, i, -1, -i\}$ under multiplication. We can define a function $f : Z_4 \to \{1, i, -1, -i\}$ by:

$$f(0) = 1, \quad f(1) = i, \quad f(2) = -1, \quad f(3) = -i$$

This function preserves the group operation, since for any $a, b \in Z_4$, we have:

$$f(a+b) = f((a+b) \mod 4)$$

and

$$f(a) \cdot f(b) = f(a) \cdot f(bmod4)$$

Now, we note that the function f is a bijection, since it is both injective and surjective. Therefore, we can conclude that the group Z_4 under addition modulo 4 is isomorphic to the group of complex numbers $\{1, i, -1, -i\}$ under multiplication. This illustrates how two seemingly different groups can have the same structure and be considered the same in terms of group theory.
Note also that the function in the previous example is not a bijection and hence does not have an inverse, however, it still preserves the group operation. This tells us that groups can have similar structures even if they are not isomorphic, and that the concept of homomorphisms allows us to compare groups in a more general way than just looking for isomorphisms.

Definition 1.7.0.1

Group Homomorphism
Let $(G, \star), (H, \cdot)$ be two groups. A group homomorphism from $G \to H$ is a map $f : G \to H$ such that for all $g_1, g_2 \in G$, we have:

$$f(g_1 \star g_2) = f(g_1) \cdot f(g_2)$$

Note 1.7.0.1

We note the following points -

- For f defined as above, we have $f(e_G) = e_H$, where e_G and e_H are the identity elements of G and H respectively. This can be shown as follows:

$$f(e_G) = f(e_G \star e_G) = f(e_G) \cdot f(e_G)$$

  Let $f(e_G) = s$. Then we have $s = s \cdot s$. Multiplying both sides by s^{-1} gives us $e_H = s$, which means $f(e_G) = e_H$.

- For any $g \in G$, we have $f(g^{-1}) = f(g)^{-1}$. This can be shown as follows:

$$e_H = f(e_G) = f(g \star g^{-1}) = f(g) \cdot f(g^{-1})$$

  Multiplying both sides by $f(g)^{-1}$ gives us $f(g) \cdot f(g)^{-1} = e_H$, which means $f(g^{-1}) = f(g)^{-1}$.